2/8/26

Experiment - 4

Aim:
To design, setup, and configure a Local Area Network (LAN) connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource / data sharing across the network.

Procedure:

1. Open cisco Packet Tracer
2. From the device category panel, select End Devices & drag 4 PCs onto the Workspace.
3. Select Switches and drag 1 Switch (eg: 2960 Series) onto the Workspace.
4. Click Connections → Select Copper Straight - Through Cable.
5. Click each PC, then click the switch and select an available FastEthernet port to completete each connection
6. Repeat all four PC's. Green link lights confirm a successful physical connection.
7. Click a PC → Desktop tab → IP Configuration
8. Enter the static IP address & Subnet Mask as per the IP table for each PC.
9. Click the Switch → CLI tab.
10. Enter the following Commands to assign a management IP:

```
Enable
Configure terminal
interface Vlan 1
ip address 10.1.1.5.255.0.0.0
no shutdown
```

exit.

11 click a PC → Desktop tab → command prompt

12. Use the ping command to test connectivity to other PC's eg, ping 10.1.1.2.

13 Successful "Reply from ..." messages confirm connectivity

14. Add a Server device (End Devices → Server-PT) and connect it to Switch.

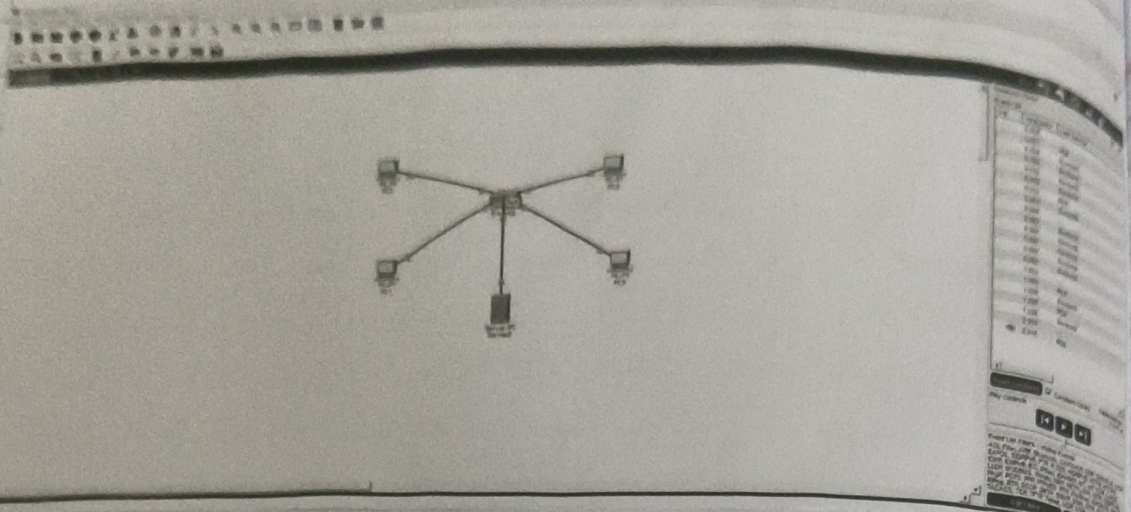
15 Assign it an IP address (eg 10.1.1.10 subnet mask) via its Desktop → IP configuration tab.

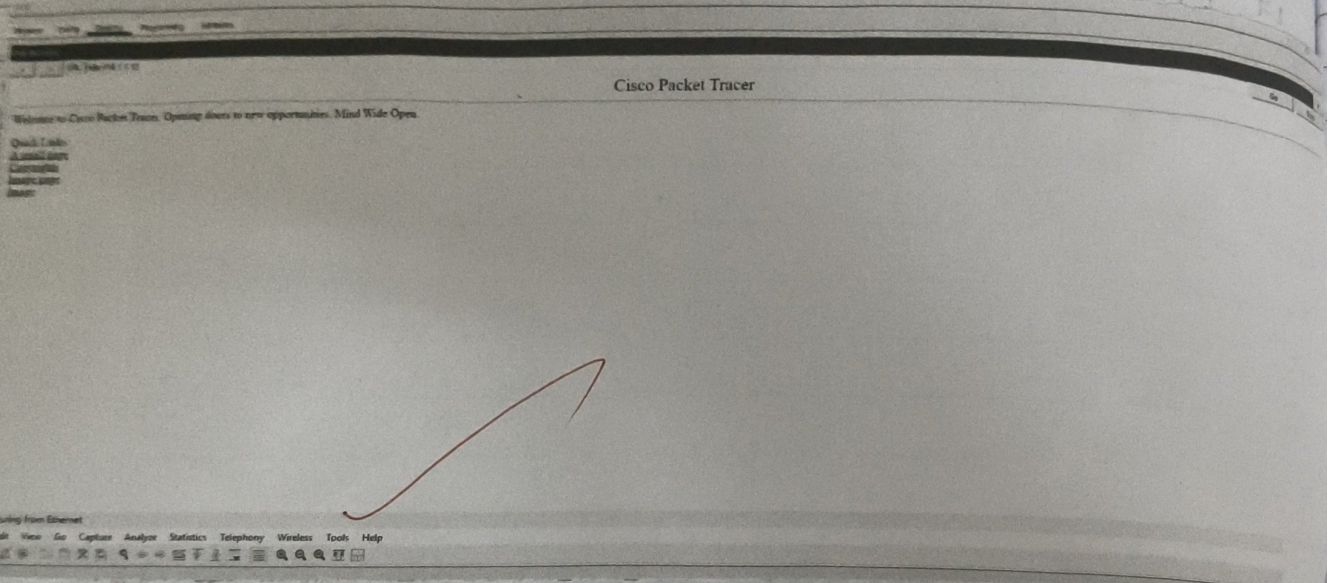
16. Open the Server's Services tab and enable the HTTP Service.

17. From another PC, open the Web Browser and enter the Server's IP address to load the page, confirming successful data transfer across the LAN

18. Switch to Simulation mode.

19. Send a ping or Web request between two devices and use the play / Capture controls to step through the packet as it travels hop-by-hop through the Switch, observing MAC address-based forwarding.





Result:

Thus the design, setup, configure a LAN and has successfully verified connectivity using ping command and demonstrated resource / data sharing across the network.